

Rogers-Ramanujan Derivation — cycles 13-15

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Rogers-Ramanujan Derivation — cycles 13-15

Introduction

Cycles 13-15 were closure cycles. They did not add new proof steps, scripts, figures, data files, or documentation. Their purpose was to confirm that the long-exposure Rogers-Ramanujan derivation was already complete and that no concrete defect had appeared that would justify reopening the proof search.

The directive was to derive the two Rogers-Ramanujan identities in a formal power series setting:

$$\sum_{n \geq 0} \frac{q^{n^2}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+1})(1 - q^{5m+4})},$$

and

$$\sum_{n \geq 0} \frac{q^{n^2+n}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+2})(1 - q^{5m+3})}.$$

The completed proof route, already validated before these cycles, is:

1. formal power series setup and truncation rules;
2. finite q-binomial and Jacobi-type product algebra;
3. an internally derived triangular matrix transform;
4. explicit alpha sequences for the two Rogers-Ramanujan cases;
5. coefficientwise limiting in the formal setting;
6. modulo-5 product cancellation;
7. final synthesis into the two identities.

Cycles 13-15 repeatedly recorded the same decision: the primary directive is complete, and the run should not continue unless a specific proof defect is found.

Definitions and Notation

The proof works in the ring $\mathbb{Z}[[q]]$, the ring of formal power series with integer coefficients. Equality in this ring means coefficientwise equality: two series are equal when every coefficient of q^N agrees.

The q-Pochhammer symbol used throughout the run is

$$(q; q)_n = \prod_{k=1}^n (1 - q^k).$$

Infinite products are interpreted formally by coefficientwise stabilization. For any fixed coefficient degree N , only finitely many product factors can affect the coefficient of q^N , so the coefficient is well-defined.

The project tracked proof progress by milestones:

milestone	meaning	current status
M0	formal setup and notation	validated
M1	finite experiment harness	validated
M2	transfer mechanism from series to product structure	validated
M3	product-side formalization	validated
M4	final proof synthesis and validation separation	validated

These milestone statuses are recorded in `promise_ledger.jsonl`, `docs/validation.md`, and `docs/lemmas/lemma_catalogue.md`.

Results

Theorem 1: First Rogers-Ramanujan Identity

The first Rogers-Ramanujan identity is recorded as proved in the final proof document:

$$\sum_{n \geq 0} \frac{q^{n^2}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+1})(1 - q^{5m+4})}.$$

Cycles 13-15 did not modify this theorem. They confirmed that it remains part of the validated M4 proof chain. The proof depends on the derived triangular transform, the $a = 1$ alpha sequence, the coefficientwise limiting transform, and formal cancellation of the modulo-5 product factors.

Theorem 2: Second Rogers-Ramanujan Identity

The second Rogers-Ramanujan identity is also recorded as proved in the final proof document:

$$\sum_{n \geq 0} \frac{q^{n^2+n}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+2})(1 - q^{5m+3})}.$$

Cycles 13-15 did not modify this theorem. They confirmed that it remains part of the validated M4 proof chain. The proof depends on the same internally derived transform, with the $a = q$ specialization and the corresponding modulo-5 product cancellation.

Cycle 13 Closure

Cycle 13 consisted of three records:

role	session ID	content
researcher	80fa1bf4-9873-4be4-9a50-8fcc7955284d	Stated that the directive was complete, with M0-M4 validated and the final proof in docs/proof/final_proof.md.
worker	f74f6f7d-b469-4a2e-b12e-cb86a8e51b9d	Boolt nothing, ran no commands, and reported that the durable proof artifacts were already in place.
auditor	1ab98e96-cda1-465b-bd39-9bcd9f44b44b	Validated closure, with no critical or moderate findings.

The cycle 13 auditor recorded only standing minor warnings: `promise_check` reported known orphan-artifact warnings for session database files and cycle reports, and `org_check` reported the known root-file warning for `rogers_ramanujan_run_config.yaml`.

Cycle 14 Closure

Cycle 14 repeated the closure decision:

role	session ID	content
researcher	776a5f4d-1615-43bd-a3af-b99fedca384d	Stated that no active questions remained and no proof search, build, run, or documentation change was recommended.
worker	4d158a72-00e3-4746-a83f-c4c5a9c53b29	Boolt no artifact and ran no commands.
auditor	5213f248-4ddb-47c9-b426-33e3b58445a8	Validated closure again, with no critical or moderate findings.

The cycle 14 rationale was that the ledger already showed full M0-M4 validation, the worker introduced no new claims, and validator output contained only known minor organizational warnings.

Cycle 15 Closure

Cycle 15 was the final closure confirmation in this range:

role	session ID	content
researcher	20d61278-9ccf-4a61-abaf-f9b5ccb64bc8	Recorded final closure and recommended no further work.
worker	03a39928-17b6-4155-873c-100402fccb0d	Boolt nothing, ran no commands, and reported that all directive success criteria were met.
auditor	72be0093-1bcb-4ba6-b87e-808a4b35af28	Validated the closure cycle and repeated the instruction not to reopen the run without a concrete proof defect.

The cycle 15 audit report is the report supplied for this cycle range. It states that no artifacts were created or modified, M0-M4 remain validated, and the durable records are present.

Remarks

The closure rule established in these cycles is explicit: the Rogers-Ramanujan proof campaign should not be reopened unless a concrete defect is identified. Examples named in the audit records include an invalid cancellation, missing lemma proof, exponent mismatch, limiting error, or residue-class mismatch.

The direct partition-bijection route remains outside the completed primary directive. Earlier cycles explored and rejected several simple bijection candidates. Cycles 13-15 did not reopen that extension.

The computational artifacts remain classified as discovery and regression support. The final proof does not depend on coefficient checks as proof substitutes.

Open Questions

No open question remains for the primary directive.

A future optional extension could pursue a direct partition bijection between gap-two partitions and the two modulo-5 residue classes. That would be a new objective, not unfinished work from the validated proof campaign.

References

No external references are cited in this report. No REFERENCES.md file exists in the workspace.

Appendix: Implementation Details

Durable Proof Artifacts

The closure cycles identify the following files as the durable proof record:

file	role
docs/proof/final_proof.md	Final self-contained proof of both Rogers-Ramanujan identities.
docs/proof/bailey_matrix_transform.md	Derived triangular transform, explicit alpha pairs, and coefficientwise limiting transform.
docs/proof/product_side.md	Formal modulo-5 product cancellation.
docs/lemmas/lemma_catalogue.md	Catalogue of definitions, proved lemmas, rejected candidates, and final theorems L41-L42.
docs/validation.md	Separation of proof-bearing results from discovery and regression computations.
promise_ledger.jsonl	Milestone and audit ledger for M0-M4.
MANIFEST.md	Current implementation inventory and cross-reference snapshot.

Current Inventory

The current manifest records:

category	count	lines
Wolfram scripts	10	1405
Python scripts	10	1166
Total scripts	20	2571
Proof/lemma/validation documents	13	3036
Experiment data and figure files under data/finite_experiments	82	n/a

No manifest update was needed for cycles 13-15 because the cycle records state that no artifacts were created or modified.

Figure Inventory

No new figures were produced during cycles 13-15. Existing figures retained from earlier cycles are:

figure	role
data/finite_experiments/rr_difference_heatmap.png	Finite coefficient agreement and negative-control diagnostics.
data/finite_experiments/q_difference_residuals.png	Series-side q-difference residual diagnostics.
data/finite_experiments/product_transform_residuals.png	Product-transform candidate residuals.
data/finite_experiments/transformed_cancellation_signals.png	Survivor patterns.
data/finite_experiments/nonlocal_involution_fixed_point.png	Nonlocal transfer fixed-point diagnostics.
data/finite_experiments/mod5_state_support.png	Modulo-5 shifted-state support diagnostics.
data/finite_experiments/euler_tail_lattice_residuals.png	Euler tail lattice residual diagnostics.
data/finite_experiments/direct_bijection_abacus_examples.png	Simple partition-bijection abacus examples.
data/finite_experiments/bailey_transform_residuals.png	Bailey-style transform residual checks.

These figures are not embedded in the main report body because cycles 13-15 produced no new visual evidence.

Validation Status

The supplied audit report for cycles 13-15 records:

severity	findings
critical	none
moderate	none
minor	known <code>promise_check</code> orphan-artifact warnings and known <code>org_check</code> root-file warning

The audit decision is **VALIDATED**.

Session Cross-Reference Map

cycle	researcher	worker	auditor	conclusion
13	80fa1bf4-9873-4be4-9a5d-f67c-79452ba2e-b12e98185a8a51-c65b-bd39-9b-cdd9f44ed4c			no new work.
14	776a5f4d-1615-43bd-a21f-5a97e-d0a3-747146-a52f3f245a9d5b-529c9-b426-88-3d5844eda8			no new work.
15	20d61278-9ccf-4a61-a0a3-f925-c1764-c155-872e-1004402fc-940a6-b37e-8d8a4b35af28			validated; no new work.

Source Inventory

The report draws from:

source	use in report
Supplied directive	Defines the target identities and proof constraints.
Supplied audit report	Provides the cycle-range validation decision.
Supplied cycle session IDs	Establishes the cycle 13-15 closure chronology.
data/sessions.db	Full session records for all nine cycle IDs.
plan_of_record.md	Defines goals G1-G3 and milestones M0-M4.
MANIFEST.md	Provides script inventory, file counts, and cross-reference map.

source	use in report
<code>promise_ledger.jsonl</code>	Records milestone validation events, including M4 validation.
<code>docs/proof/final_proof.md</code>	Durable final proof artifact.
<code>docs/lemmas/lemma_catalogue.md</code>	Lemma and theorem status catalogue.
<code>docs/validation.md</code>	Validation separation between proof and computation.